

Background of the Study

The Sangguniang Kabataan (SK) plays a crucial role in promoting youth engagement by implementing programs and activities that contribute to the development and participation of young people in the community. Its responsibility is to create and implement projects directed toward the youth, including initiatives related to sports, education, and health. In Barangay Concepcion Dos, Marikina City, SK officials regularly organize activities such as sports events, scholarship incentives, school supply assistance, and community development programs. However, despite these efforts, SK officials encounter several operational challenges, particularly in the manual documentation of records, the repetitive preparation of similar documents for different activities and events, the management of youth participation, and the tracking of project expenditures. The study aims to develop a system that integrates an AI-powered chatbot with modules for event management, participation tracking, and AI-driven project budget estimation to assist SK officials in managing youth programs and activities more efficiently. The AI chatbot is designed to provide automated assistance in responding to SK inquiries, guiding users through SK services, and supporting administrative tasks. In addition, the event management and participation tracking modules will allow SK officials to digitally manage registrations, attendance records, and participants information, reducing reliance on manual documentation and improving the efficiency of organizing youth-related activities.

The system will also incorporate an AI-driven project budget estimation feature that analyzes historical activity data and existing templates to provide recommendations

for planning future SK projects and events. To support contextual decision-making, the system will utilize localized datasets for pattern recognition and analysis of youth activities within the barangay. Furthermore, the system will be deployed on a dedicated local server to ensure data security, system reliability, and accessibility even in areas with limited internet connection. Through these integrated features, the proposed system aims to improve administrative efficiency, enhance transparency in SK operations, streamline documentation processes, and ultimately strengthen youth engagement and service delivery within Barangay Concepcion Dos.

Objectives of the Study

The general objective of the study is to develop a web-based administrative framework that improves the efficiency of managing youth programs, documentation, and public services for the Sangguniang Kabataan of Barangay Concepcion Dos through the integration of artificial intelligence.

Specifically, the study aims:

1. To automate the preparation of official SK compliance documents and project resolutions to mitigate the repetitive and high-volume manual administrative workflow.
2. To provide an AI-driven project budget estimation feature that analyzes historical expenditure patterns to assist in financial planning and detect potential budget imbalances.

3. To establish a digital record-keeping and participation tracking system that replaces manual registration and secures administrative reports in a centralized environment.
4. To develop an AI-powered chatbot that provides automated assistance in responding to SK inquiries and guiding users through administrative services.
5. To use the ISO-25010 quality standard to measure and evaluate the design, performance, and usability of the aSK Youth system.

Scope and Limitation of the Study

The scope of the study is to develop the "aSK Youth: An AI-Driven SK Management and Public Service System" to enhance the administrative efficiency and data reliability of Sangguniang Kabataan operations in Barangay Concepcion Dos.

- The proposed study will automate the generation of various administrative records and reports required for SK project implementation and official compliance.
- The proposed study will utilize a localized dataset of historical records and RA 10742 to provide context-aware AI budget recommendations.
- The proposed study will incorporate a digital verification system for monitoring registrations and attendance during SK-led youth events.
- The proposed study will develop a web-based architecture using React.js, Node.js, Python and a large language model (LLM) that is optimized for mobile responsiveness.

- The proposed study will offer AI-driven chatbot assistance to streamline internal SK communications and information dissemination.

The limitations of the study which are not included in the study are the following:

- The proposed study will not include administrative functionality for areas beyond Barangay Concepcion Dos.
- The proposed study will not involve integration with other local government units (LGUs) or non-SK operations.
- The proposed study will not incorporate advanced features such as biometric verification, multi-language support, or real-time video.
- The proposed study will not guarantee 100% AI accuracy if there is a lack of quality or completeness in the historical SK data provided.
- The proposed study will require a stable internet connection for the AI modules once transitioned to web hosting.
- The proposed study will require a computing device with sufficient hardware specifications to support the local AI inference modules, including a dedicated GPU with at least 6GB of VRAM to run locally.

Significance of the Study

A discussion of the significance of the study for ASK Youth: An AI-Driven SK Management and Public System for Barangay Concepcion Dos highlight the importance of the system, the individuals and groups who will benefit once it is implemented, its potential advantages, and its overall impact on youth governance and community engagement.

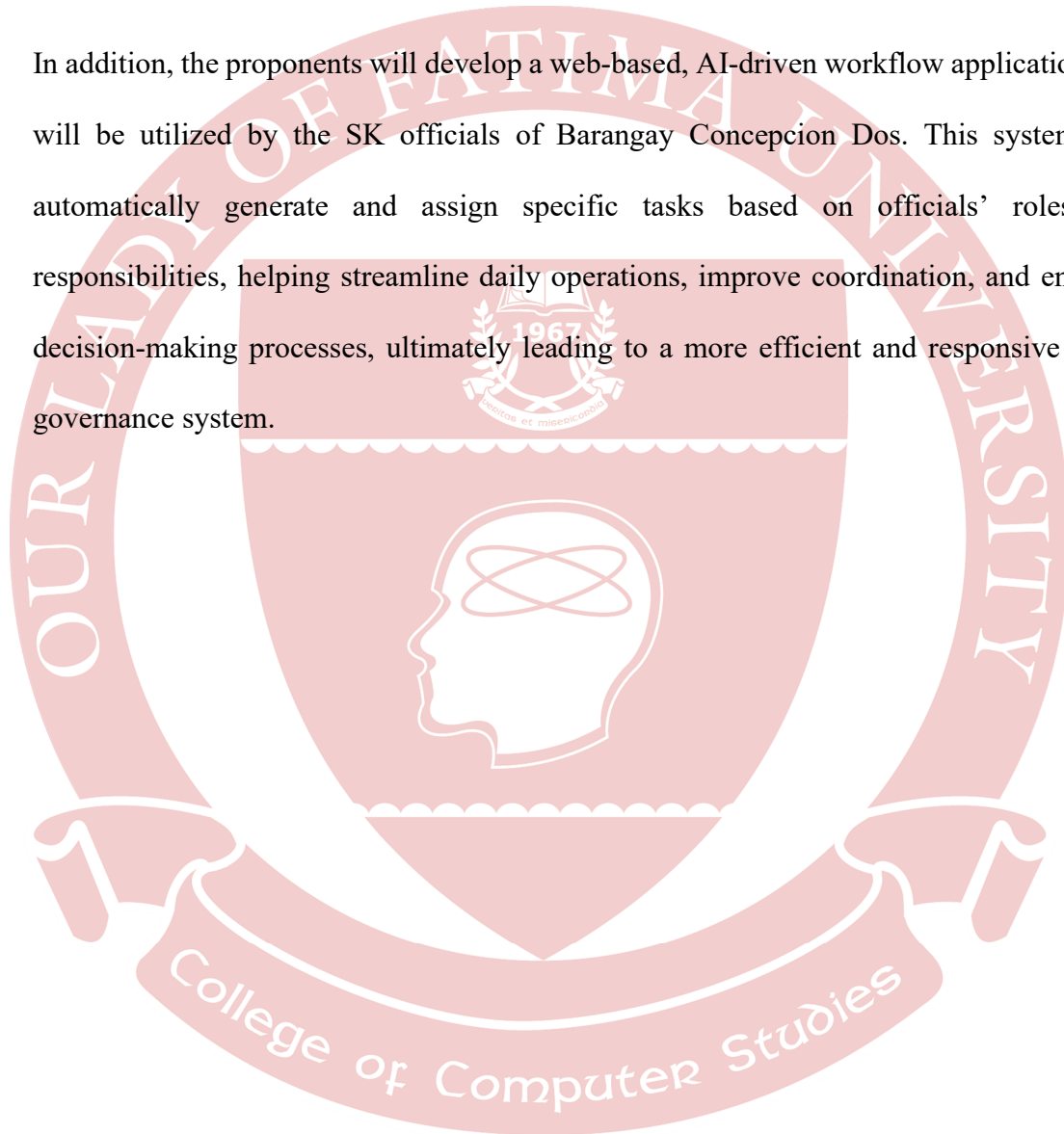
SK Officials. This study will benefit SK officials by providing an AI-driven workflow application that automates task generation and assignment based on their roles and responsibilities. The system helps streamline daily operations, improve coordination, and ensure timely completion of activities. Its AI capabilities support decision-making, enhance planning and organization, and promote more efficient and effective youth governance.

Community. This study will benefit the community by improving the efficiency and organization of SK officials through an AI-driven workflow application that generates and assigns specific tasks. By streamlining operations and enhancing coordination, the system enables the delivery of more organized programs and timely services, resulting in a more responsive, well-managed, and connected barangay.

Researchers. This study will benefit researchers by providing knowledge and experience in developing an AI-driven management system.

Future Researchers. This study may serve as a reference for future researchers who wish to improve or expand similar systems.

In addition, the proponents will develop a web-based, AI-driven workflow application that will be utilized by the SK officials of Barangay Concepcion Dos. This system will automatically generate and assign specific tasks based on officials' roles and responsibilities, helping streamline daily operations, improve coordination, and enhance decision-making processes, ultimately leading to a more efficient and responsive youth governance system.



Foreign Studies

The following references, including authors, articles, books, and journals, represent foreign studies and their corresponding relevance to the proposed study.

AI Integration in Organizational Workflows

According to Oliveira et al. (2025), artificial intelligence (AI) has rapidly expanded due to advancements in data storage, computing capabilities, and improved methodologies in machine learning, big data analytics, and robotics. This widespread adoption has significantly influenced the way people work, as AI systems are now capable of performing tasks that were once exclusively handled by humans. As a result, AI is transforming organizational processes by automating routine activities, enhancing productivity, and reshaping how tasks and workflows are managed. These developments highlight the growing importance of AI-driven systems in improving efficiency and supporting decision-making in various sectors, including public administration. Retrieved from <https://www.mdpi.com/2078-2489/16/9/764?>

With regard to the study of Oliveira et al. (2025), it shares a similar objective with the present study in improving organizational efficiency through the use of artificial intelligence. The study emphasizes how AI transforms work processes by automating routine tasks, thereby increasing productivity and reducing workload. Similarly, the proposed study aims to enhance the current workflow of SK officials, particularly in task

management, coordination, and planning. Existing manual processes often result in delays, miscommunication, and disorganized execution of activities. In contrast, Oliveira et al.'s study demonstrates how AI can restructure workflows by automating both routine and cognitive tasks, enabling users to focus on more critical responsibilities.

AI Integration in Malaysian Public Administration for Improved Governance

According to Firdaus et al. (2024), the rapid advancement of technology, particularly artificial intelligence (AI) and digitalization, has significantly influenced public administration across Southeast Asia, including Malaysia. AI-driven systems have been widely adopted to optimize workflows, automate routine tasks, and improve overall efficiency. By integrating these tools into administrative operations, governments have been able to streamline task management, reduce human error, and enhance decision-making through data-driven insights. Retrieved from <https://rsisinternational.org/journals/ijriss/articles/ai-integration-in-malaysian-public-administration-for-improved-governance/?>

In accordance with Firdaus et al. (2024), this study similarly aims to improve workflow efficiency using AI. The research highlights that AI can automate repetitive and cognitive

tasks, reduce manual workload and minimizing errors. By reorganizing administrative processes, AI-driven systems streamline task allocation, improve coordination, and ensure that responsibilities are executed systematically and effectively. Furthermore, AI adoption allows officials to focus on strategic tasks that require human judgment and problem-solving, rather than routine operations. Leveraging AI for workflow optimization can lead to significant efficiency gains, better task distribution, and a more organized approach to administrative responsibilities, ultimately supporting more effective and responsive governance.

The strategic use of AI in the public sector

According to Hjaltalin et al. (2024), Artificial Intelligence (AI) has developed rapidly over the past decade, and its abilities and potential have grown accordingly. AI is also gaining popularity in the public sector and is viewed as “a response to the growing complexity of the public sector and need to handle larger and more advanced forms of data” (Schou & Hjelholt, 2017). These developments give rise to new opportunities but also challenges for governments. Furthermore, AI adoption allows public officials to focus on strategic, higher-level tasks that require human judgment, such as policy planning, problem-solving, and stakeholder engagement.

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from

<https://www.sciencedirect.com/science/article/pii/S0740624X24000066?>

In accordance with Hjaltalin et al. (2024), the present study similarly aims to improve administrative efficiency and workflow management through AI. The research highlights that AI-driven systems can automate repetitive and cognitive tasks, reorganize work processes, and enhance productivity. By leveraging AI for workflow optimization, the study seeks to reduce delays, improve coordination, and ensure systematic execution of responsibilities, ultimately enabling public officials to focus on strategic tasks and decision-making that contribute to more effective and responsive governance.

Integrating AI into Public Governance

According to Aarab et al. (2025), over the past decade, artificial intelligence (AI) has progressively reshaped how public administration operates, makes decisions, and delivers services. AI has increasingly been utilized in public sector operations, ranging from traffic optimization and fraud detection to the automation of administrative tasks. These developments highlight the role of AI in workflow optimization and task automation, enabling public institutions to streamline processes, reduce manual workload, and enhance decision-making through data-driven insights. Retrieved from <https://www.mdpi.com/2673-6470/5/4/59?>

In relation to the study of Aarab et al. (2025), the present study similarly aims to enhance administrative efficiency through an AI-driven workflow system. By automating both routine and cognitive tasks, optimizing task allocation, and improving coordination, the

proposed system seeks to address common challenges such as delays, miscommunication, and disorganized processes. Through the integration of AI, the study aims to achieve greater workflow efficiency, improved task management, and more effective decision-making, ultimately contributing to more responsive and efficient governance at the local level.

Towards Smart Public Administration

According to recent studies (Chong et al., 2021; Medaglia et al., 2021), the rapid rise of artificial intelligence (AI) technologies has significantly transformed public administration, particularly in terms of service delivery, citizen engagement, and internal operations. Among these innovations, AI chatbots have emerged as effective tools for improving administrative efficiency, enabling real-time communication, and enhancing responsiveness in government services (Sousa & Rocha, 2019; Wirtz et al., 2021). However, most existing studies approach AI integration from a technology-centered perspective, focusing primarily on infrastructure, innovation capabilities, and organizational readiness, while often overlooking institutional, political, and environmental factors that influence adoption, especially in developing and transitioning countries (Aoki, 2020; Sharma et al., 2022). Retrieved from <https://www.mdpi.com/2076-3387/15/8/324?>

In relation to the present study, these findings emphasize the importance of considering both the benefits and challenges of AI integration in public administration. While AI-driven

systems can significantly enhance workflow efficiency, automate tasks, and improve decision-making, their successful implementation depends on factors such as system usability, organizational support, and user acceptance. Thus, the proposed AI-driven workflow system for SK officials aims to address these challenges by providing a user-friendly platform that automates task generation, improves coordination, and supports efficient governance at the local level.

Foreign Literature

The following articles and viewpoints presented below are based on foreign literature and their corresponding synthesis.

AI capacity in the public sector

According to Legner et al. (2017) in recent years, digital transformation efforts of public institutions have accelerated with artificial intelligence (AI) playing a crucial role. Governments are increasingly integrating digital technologies to modernize their operations. AI has been widely adopted to enhance administrative processes, improve service delivery, and increase operational efficiency (Misuraca et al., 2020). This growing adoption reflects the need for public organizations to remain adaptive in dynamic environments, where AI serves as a tool for organizational flexibility and responsiveness (Ayinaddis, 2025).

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With reference to the studies of Legner et al. (2017), Misuraca et al. (2020), and Ayinaddis (2025), the proponents share the same goal of using artificial intelligence to make public services faster and more efficient. These studies support the aSK Youth system by showing how AI can help improve workflows, provide better services, and make communication easier between SK officials and the youth.

AI adoption and diffusion in public administration

Artificial intelligence (AI) has emerged as a key driver of technological innovation in public administration, building on earlier e-government efforts aimed at improving efficiency and reducing costs. According to Madan et al. (2022), AI is expected to significantly influence not only government employees but also citizens and society as a whole. As AI begins to play a role in public decision-making, it brings major changes to how government functions and raises important issues related to power, trust, and legitimacy. These concerns highlight the growing impact of AI on people's lives. Therefore, understanding the different factors that affect the adoption and use of AI is essential in ensuring that it creates value in public administration. Retrieved from https://www.sciencedirect.com/science/article/pii/S0740624X22001101?fr=RR-2&ref=pdf_download&rr=9e55211ad8de6e64

With reference to the study of Madan et al. (2022), proponents share the same goal of utilizing artificial intelligence to improve public service systems and enhance citizen engagement. The findings of this study support the development of the ASK Youth system, particularly in applying AI to streamline administrative workflows, assist in decision-making, and improve communication between SK officials and youth constituents. Furthermore, the study emphasizes the importance of trust, transparency, and proper implementation, which are essential considerations in ensuring that the proposed system is effective, reliable, and beneficial to the community.

The Rise of Artificial Intelligence (AI) in Public Administration

According to Fibriyanita, F. (2024), the integration of artificial intelligence (AI) in public administration has significantly transformed the delivery of government services and internal operations. AI technologies, which enable machines to perform cognitive tasks such as learning, problem-solving, and decision-making, are increasingly being adopted worldwide to enhance efficiency and service quality (Katekar, 2023). Governments in countries such as Indonesia and India have utilized AI to improve service delivery and crisis management, demonstrating its ability to create public value through automation and data-driven decision-making (Yusriadi et al., 2023; Katekar, 2023). Moreover, AI plays a crucial role in improving public services by addressing challenges such as inefficiency, bureaucratic delay, and lack of transparency (Dwivedi et al., 2021). In sectors such as healthcare, education, and urban management, AI systems can process large volumes of

data, identify patterns, and support faster and more accurate decision-making (Robles & Mallinson, 2023). Retrieved from <https://journal.ppipbr.com/index.php/pusba/article/view/272>

The proponents' study and the study of Fibriyanita (2024) share the same goal of improving efficiency and service delivery in public administration using artificial intelligence. However, while Fibriyanita's study focuses on the general application of AI in enhancing public services, the proponents' study specifically aims to develop an AI-driven workflow system for SK officials. The proposed system will utilize AI to automate routine tasks, improve task coordination, and organize administrative processes more effectively. Furthermore, the system is designed to address common issues such as delays, miscommunication, and unstructured workflows. By implementing this solution, the proponents aim to create a more efficient and responsive workflow, enabling officials to manage tasks more quickly and accurately, ultimately improving overall service delivery within the barangay.

Artificial intelligence (AI) applied to public management

According to Ocaña-Fernández et al. (2022), Artificial intelligence (AI) has evolved significantly with the advancement of computing technologies. According to Chui, AI refers to the ability of machines to exhibit human-like intelligence, such as problem-solving and decision-making. Recent developments in machine learning, big data, and deep

learning have accelerated AI adoption across various sectors, including public administration. These advancements enable organizations to improve efficiency, productivity, and reliability by automating tasks and supporting decision-making processes. Furthermore, AI has demonstrated its potential to transform traditional workflows by replacing human activities in different operational contexts. Despite its benefits, concerns regarding workforce displacement and ethical considerations remain, as automation may significantly affect existing job roles. Nevertheless, AI continues to be a powerful tool that reshapes organizational processes and work environments. Retrieved from <https://aber.apacsci.com/index.php/met/article/view/1795?>

In accordance with the study of Ocaña-Fernández et al. (2022), the proponents of this study share a similar objective of utilizing artificial intelligence to improve efficiency and streamline processes. The study emphasizes how AI can automate tasks, enhance productivity, and support decision-making within organizations. Similarly, the proposed study aims to improve the workflow of SK officials by developing an AI-driven system that automatically generates and assigns specific tasks. While Ocaña-Fernández et al.'s study focuses on the broader application of AI in transforming organizational processes, the proponents' study applies this concept specifically to local governance. Both studies highlight the importance of AI in reducing manual work, improving coordination, and optimizing performance, ultimately contributing to more efficient and effective operations.

Exploring the nexus of governance and AI ethics

According to Ikram et al. (2026), recent advances in artificial intelligence (AI) have led to technologies capable of performing cognitive functions similar to the human mind (Rai et al., 2019). AI is reshaping human-technology interactions by offering more interactive and intelligent support for tasks than previous technologies (Maedche et al., 2019). Unlike traditional technologies, AI can operate autonomously (Rieder et al., 2020; Scherer, 2015). Notable examples include generative AI and large language models, which have expanded their applications across organizational contexts (Markus & Rowe, 2023). These developments have significant implications for work structures. AI can enable human-machine hybrid work, in which autonomous systems complement human decision-making and operational activities (Mollick et al., 2024). AI systems are increasingly capable of performing tasks independently, reducing direct human oversight (Baird & Maruping, 2021). This shift in self-governance, moving control from humans to AI, introduces ethical challenges such as fairness in task distribution, social exclusion, unequal treatment, and transparency concerns (Boada et al., 2021). Retrieved from <https://www.sciencedirect.com/science/article/pii/S254392512600001X?>

In relation to Ikram et al (2026), these findings demonstrate the potential of AI to enhance workflow and task management for SK officials. By implementing AI systems capable of autonomously monitoring tasks, notifying relevant stakeholders, and assisting in decision-making, the proposed system aims to reduce delays, improve coordination, and optimize

task execution. Furthermore, adherence to ethical principles ensures that the AI-driven system supports officials fairly and transparently, aligning with the study's goal of creating an efficient, reliable, and socially responsible task management tool.

Local Literature

The following articles and opinions stated below rests in the local literature and its corresponding synthesis

Web-Based Barangay Information System

According to San Jose et al. (2024), web-based barangay information systems have become essential in improving administrative efficiency in local government units. Their study emphasized the integration of digital databases and AI-driven tools to manage resident information, monitor demographic data, and streamline documentation processes. Traditional

manual systems often result in delays, data redundancy, and difficulty in retrieving records, which negatively affect service delivery.

The researchers explained that implementing a web-based system enables real-time access to

data, improves accuracy, and enhances record management. Features such as centralized databases, automated reporting, and secure user authentication contribute to efficient

governance. Additionally, AI tools allow the system to analyze patterns and provide insights for

better decision-making.

These findings highlight the importance of transitioning from manual to digital systems to improve operational efficiency. Retrieved from:

<https://ojs.aaresearchindex.com/index.php/aasgbcpjmra/article/view/15594>

In relation to the study of San Jose et al. (2024), the present study similarly aims to improve administrative efficiency through a centralized web-based system. By integrating a digital database, the proposed system enhances data management, accessibility, and transparency in SK operations. This aligns with the goal of reducing manual processes and improving service delivery in Barangay Concepcion Dos.

Event Management System for LGUs

According to Goh et al. (2021), digital event management systems improve the organization and execution of community activities in local government units. Their study addressed issues such as scheduling conflicts, miscommunication, and inefficient coordination caused by manual processes. The system developed included features such as event scheduling, participant registration, and automated notifications. These features improved coordination among organizers and participants, while real-time monitoring allowed efficient tracking of events. The study found that digital systems significantly reduce errors and improve overall efficiency in managing community programs.

Furthermore, the system enhanced transparency by maintaining digital records of events, allowing better evaluation and planning for future activities. Retrieved from: <https://paraverse.feutech.edu.ph/repository/article/10.1109/ICCIKE47802.2019.9004391>

In relation to the study of Goh et al. (2021), the present study incorporates an event management module to improve the organization of SK activities. By automating scheduling and participant management, the system addresses issues such as delays and miscommunication. This contributes to more efficient and organized youth program implementation.

Participation Tracking System

According to San Jose et al. (2024), participation tracking systems are essential in monitoring community engagement and ensuring accurate documentation. Their study identified that manual attendance tracking leads to errors, missing data, and inefficiencies. The system they developed enabled real-time tracking of participants and provided automated reports for analysis. The integration of AI tools allowed the system to identify participation trends and support decision-making processes. The study concluded that digital tracking systems improve accuracy, efficiency, and data reliability. Additionally, the system enhanced communication by providing updates and notifications to participants.

Retrieved from:

<https://ojs.aaresearchindex.com/index.php/aasgbcjpmra/article/view/15594>

In relation to the study of San Jose et al. (2024), the present study implements a participation tracking module to monitor youth engagement in SK programs. By using real-time tracking and digital records, the system improves documentation and supports data-driven decision-making, aligning with the study's findings.

AI Chatbot in Public Service

According to Cuevas et al. (2024), AI chatbots significantly improve public service delivery by providing automated and real-time responses to user inquiries. Their study developed a chatbot integrated into a barangay information system using natural language processing (NLP). The chatbot was capable of answering frequently asked questions, guiding users through services, and improving accessibility. The study found that chatbot integration reduces workload for administrative staff and improves user satisfaction through faster response times. Furthermore, the chatbot enhanced accessibility by allowing users to interact with the system anytime. Retrieved from: <https://journal.esrgroups.org/jes/article/view/2426>.

In relation to the study of Cuevas et al. (2024), the present study integrates an AI chatbot to assist users in accessing SK services. By providing automated responses and guidance, the chatbot improves efficiency and enhances user experience, supporting effective public service delivery.

AI-Based Decision Support System

According to Mercurio and Hernandez (2022), AI-based decision support systems improve planning and resource allocation through data-driven insights. Their study utilized open data and analytics to generate recommendations for community-based systems. The system analyzed historical data to identify patterns and trends, enabling more accurate decision-making. It also included visualization tools that helped users understand data more effectively. The findings showed that AI improves accuracy, efficiency, and reliability in planning processes. Additionally, the study emphasized the importance of data quality in ensuring accurate results. Retrieved from: <https://arxiv.org/abs/2201.12544>

In relation to the study of Mercurio and Hernandez (2022), the present study incorporates an AI-driven budget estimation feature. By analyzing historical data, the system aims to provide accurate cost estimates and improve planning efficiency, aligning with the principles of data-driven decision-making.

Local Studies

The following citations of the author, articles, books, and journals are the local studies and its corresponding synthesis to the proposed study.

Digital Transformation in Local Governance and Automated Record Management

Dela Cruz and Santiago (2021) developed an integrated Barangay Information Management System (BIMS) that incorporates Geographic Information System (GIS) mapping to improve administrative efficiency. The study focused on transitioning from manual, paper-based records to a centralized digital platform to address issues of data redundancy and slow document retrieval. The system was designed to handle resident profiling, record management, and automated report generation while providing a visual representation of community demographics. The findings of the study revealed that the implementation of the digital system led to a marked reduction in processing time for official documents and minimized human errors. By standardizing input formats, the researchers observed a decrease in data duplication and an increase in overall record accuracy. Furthermore, the integration of GIS provided officials with better insights into resource allocation and community needs, enhancing transparency in local governance.

Retrieved from: <https://www.ijssat.org/papers/2026/1/10412.pdf>

In relation to the study of Dela Cruz and Santiago (2021), the present study similarly aims to enhance administrative efficiency through a centralized digital platform. By transitioning from manual documentation to a structured web-based application, the proposed system seeks to resolve issues of data redundancy and slow information retrieval. The focus on real-time data access in the reviewed study supports the implementation of

the centralized digital database in the proposed SK management system for Barangay Concepcion Dos.

Quality Standards in LGU Digital Operations and Event Management

Tuazon et al. (2025) conducted a study on the evaluation of Quality Management Systems (QMS) and digital maturity in Local Government Units (LGUs). The primary objective was to assess how digital platforms streamline the planning and execution of community activities. The study highlighted the importance of moving toward digital modules that include event scheduling, participant registration, and automated task tracking to ensure service standards and operational continuity. The findings indicated that digital integration significantly improved the organization of local events by reducing scheduling conflicts and minimizing manual errors in data entry. One of the key results was the improvement in accuracy for managing large volumes of event-related data, as digital systems allow for instant updates and consolidated tracking, eliminating the need for manual data reconciliation which often causes delays in community service delivery.

Retrieved from: <https://journal.ijprse.com/index.php/ijprse/article/view/1207>

In relation to the study of Tuazon et al. (2025), the present study similarly incorporates an event management module to efficiently organize and schedule youth programs. By adopting a structured approach to event coordination, the proposed system aims to address the operational challenges of manual document preparation for different activities. The integration of automated tracking features in the reviewed study validates the need for a

digital solution to improve the efficiency of organizing youth-led activities in Barangay Concepcion Dos.

Electronic Attendance Systems for Public Accountability

Garcia and Rivera (2024) developed an Electronic Attendance Monitoring System designed to solve the inefficiencies of manual tracking in community-based programs. The study aimed to address missing data and the difficulty of evaluating the impact of local initiatives. The system allowed for digital recording of attendance and was linked to a centralized database for real-time monitoring and accountability of participant engagement. The findings showed that the system drastically improved the accuracy of attendance records by eliminating duplicate entries and providing a reliable digital trail. The real-time tracking capability enabled organizers to assess participant engagement levels instantly, allowing for more informed decision-making. Additionally, the study concluded that electronic systems act as effective tools in ensuring a disciplined environment and fostering a culture of accountability among community participants.

Retrieved from: <https://iarjset.com/wp-content/uploads/2025/05/IARJSET.2025.125161.pdf>

In relation to the study of Garcia and Rivera (2024), the present study similarly implements a participation tracking system to record and monitor youth attendance in SK activities. By replacing manual logging with a digital attendance record, the proposed system seeks to improve documentation accuracy and reduce reliance on paper-based methods. The use of

real-time monitoring to evaluate program success aligns with the proposed study's objective of strengthening youth engagement and service delivery.

Utilization of AI Chatbots in Enhancing Public Service Efficiency

Villanueva et al. (2023) introduced an AI-powered chatbot system designed to streamline public service inquiries and information dissemination. The study sought to address the heavy workload of administrative personnel who were often overwhelmed by repetitive queries from citizens. The chatbot utilized natural language processing (NLP) to understand and respond to user questions in real time, providing 24/7 assistance without human intervention. The findings of the study indicated that the AI chatbot significantly enhanced user experience by providing instant responses and reducing the need for in-person visits for simple inquiries. Meta-analysis of the results showed a significant positive effect on the efficiency of public services and a reduction in response time. Users reported high levels of satisfaction with the system's ease of use and the convenience of getting quick answers to their concerns. Retrieved from: <https://journalsocialsciences.com/index.php/oaijss/article/download/255/497> In relation to the study of Villanueva et al. (2023), the present study similarly aims to develop an AI-powered chatbot to provide automated assistance for SK inquiries and public service access. By utilizing natural language interaction, the proposed system seeks to guide users through SK services more efficiently. The integration of an automated assistant in the

proposed study is intended to support administrative tasks and provide a more responsive communication channel for the youth of Barangay Concepcion Dos.

AI-Based Decision Support for Accurate Project and Budget Planning

Obanor, Mendoza, et al. (2025) examined how AI-based Decision Support Systems (DSS) increase efficiency and accuracy in project management. The study focused on moving away from manual calculations and guesswork in project planning. The system utilized AI to analyze patterns and provide immediate data feedback for resource decisions, time control, and budget allocation for small-scale community projects. The findings revealed that AI-based systems are more accurate in judging risks and monitoring activities, which decreases human errors and results in more dependable project outcomes. The study highlighted that automating repetitive jobs and giving immediate feedback makes AI-based DSS highly useful for increasing both the speed and reliability of project management tasks. Leveraging these systems allows organizations to allocate resources more effectively and ensures greater accountability. Retrieved from: https://www.researchgate.net/publication/398724048_AI-Based_Decision_Support_Systems_in_Project_Management_Enhancing_Efficiency_and_Accuracy

In relation to the study of Obanor, Mendoza, et al. (2025), the present study similarly utilizes AI-driven project budget estimation based on historical activity data. By incorporating pattern recognition and localized datasets, the proposed system aims to

provide accurate recommendations for planning future SK projects. This feature is designed to address the operational challenge of manual expenditure tracking and improve the transparency of SK financial operations.

Technical Background

The study utilizes a multi-layered software architecture consisting of React.js (v18.0) for the frontend and Node.js (v20.x) for server-side logic, integrated with a localized Large Language Model (LLM). React.js was selected for its superior component-based rendering, which is essential for maintaining a highly responsive mobile interface for SK officials. Instead of relying on traditional, monolithic API structures, the system leverages this integrated LLM to provide the specialized intelligence required for administrative tasks such as document automation and context-aware querying. The architecture is specifically optimized to run efficiently on accessible, mid-tier local hardware, eliminating the reliance on expensive enterprise cloud hosting or external API subscriptions. Tailwind CSS is utilized for UI design to ensure minimal load times on mobile devices, while VS Code serves as the primary IDE for developing the automated workflows and localized AI interactions.

Regarding hardware resources, the system follows a hardware-agnostic, web-based delivery model for the end-user, ensuring the framework is universally accessible via standard Android smartphones or tablets. To maximize flexibility and cost-efficiency for the Barangay, the system bypasses standard shared web-hosting environments—which

cannot sustain LLM workloads—in favor of self-hosting via secure internet tunneling. This tunneling method allows the beneficiary to utilize an existing local PC to expose the system to the internet, effectively deploying it as a fully functioning live web application without incurring recurring external server fees. This configuration is highly viable because the localized AI model is optimized to require significantly less capacity than larger enterprise alternatives, allowing the backend to operate stably on machines with 16GB of RAM. While the core data engineering and backend hosting are processed on a development machine equipped with an RTX 4060 Ti GPU, the final implementation remains entirely hardware-agnostic for the end-user. This enables SK officials to execute advanced, AI-driven administrative functions from their existing mobile devices without necessitating high-end equipment investments from the Barangay.

CHAPTER III

Research Methodology



This chapter presents the comprehensive research methods and systematic procedures utilized to successfully achieve the objectives of the study. It details the dual-method research design that combines descriptive-comparative evaluation and Agile-RAD iterative system development methodology. Furthermore, it defines the targeted respondents consisting of SK officials and youth constituents, the data gathering instruments including surveys and interviews, and the statistical treatment involving ISO 25010 metrics, mean, standard deviation, and t-tests.

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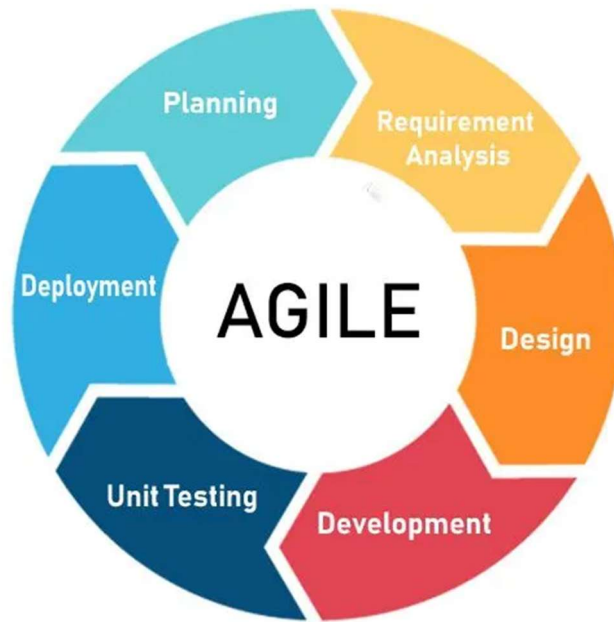


Figure 1. System Development Life Cycle

This study employs developmental and evaluative research design. The developmental phase focuses on creating the aSK Youth system using the Agile methodology. The evaluative phase uses the ISO/IEC 25010 quality standard to assess the system's quality characteristics after development.

Research Design and Approach

This study employs a developmental and evaluative research design. The developmental phase focuses on creating the aSK Youth system using the Agile methodology. The evaluative phase uses the ISO/IEC 25010 quality standard to assess the system's quality characteristics after development.

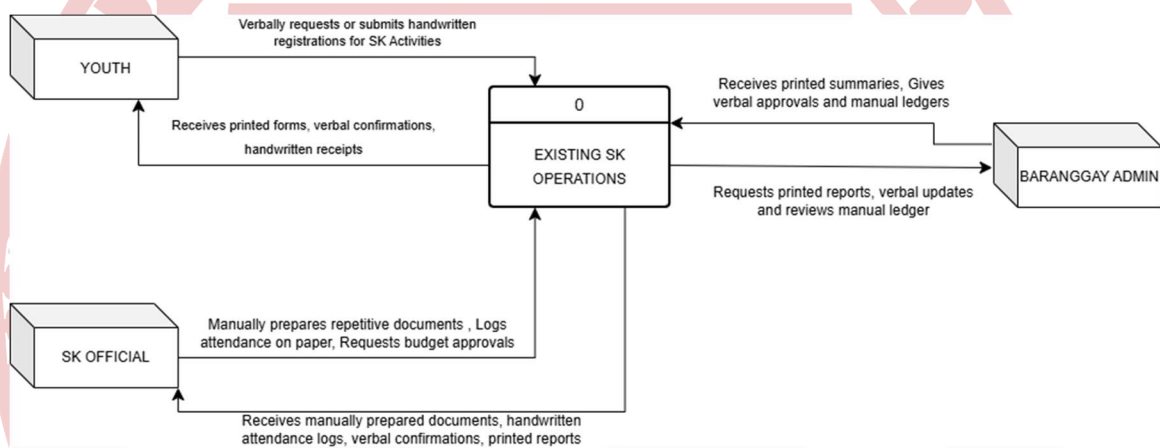


Figure 2. Existing Context Diagram (Manual SK Process)

The existing context diagram illustrates the Manual SK Operations as a paper-based system with no central digital database. The external entities which are the Youth/Resident, SK Officer, and Barangay Admin interact with the system using manual methods. Youth/Residents engage through verbal inquiries, paper announcements, printed forms, and handwritten receipts. SK Officers receive and handle requests, produce printed reports, and give verbal confirmations. Barangay Admins exchange documents, summaries, printed ledgers, and verbal updates while also making manual report requests. The diagram

highlights the fragmented, non-digital nature of current SK operations, where data flows through physical documents and verbal communication rather than a centralized electronic system.

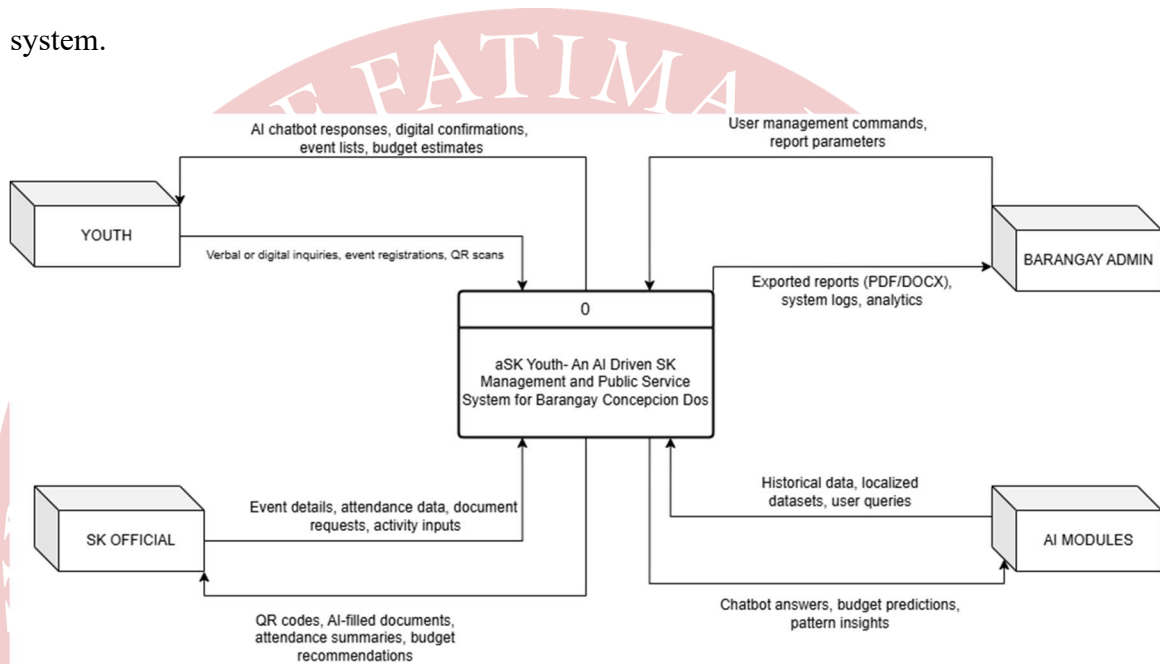


Figure 3. Proposed Context Diagram

The proposed context diagram illustrates the aSK Youth system as an AI-driven, centralized platform designed to replace manual SK operations. The external entities which are the Youth, SK Officer, and Barangay Admin interact with the system using digital and automated methods. Youth engage through verbal or digital inquiries, event registrations, and QR scans, and in turn receive AI chatbot responses, digital confirmations, event lists, and budget estimates. SK Officers provide event details, attendance data, document requests, and activity inputs, and receive QR codes, AI-filled documents, attendance summaries, and budget recommendations. Barangay Admins send user management

commands and report parameters, and receive exported reports in PDF or DOCX format, system logs, and analytics. The diagram also includes AI Modules that exchange historical data, localized datasets, and user queries with the system, returning chatbot answers, budget predictions, and pattern insights. The diagram highlights the transformed, digital nature of the proposed SK operations, where data flows through automated processes, AI-driven assistance, and a centralized electronic system rather than physical documents and verbal communication.

Data Flow Diagram

A Data Flow Diagram (DFD) is a visual tool that shows how information moves through a system or process. It uses simple shapes like rectangles, circles, and arrows along with short labels to display where data comes from, where it goes, where it is stored, and the paths it takes. DFDs can be simple, like a hand-drawn overview, or more detailed with multiple layers that go deeper into how data is handled. They are useful for analyzing an existing system or planning a new one. A DFD can visually explain things that are hard to put into words, and it is easy to understand for both technical and non-technical people, from programmers to managers. This is why DFDs remain popular. However, they are less effective today for visualizing interactive, real-time, or database-focused systems.

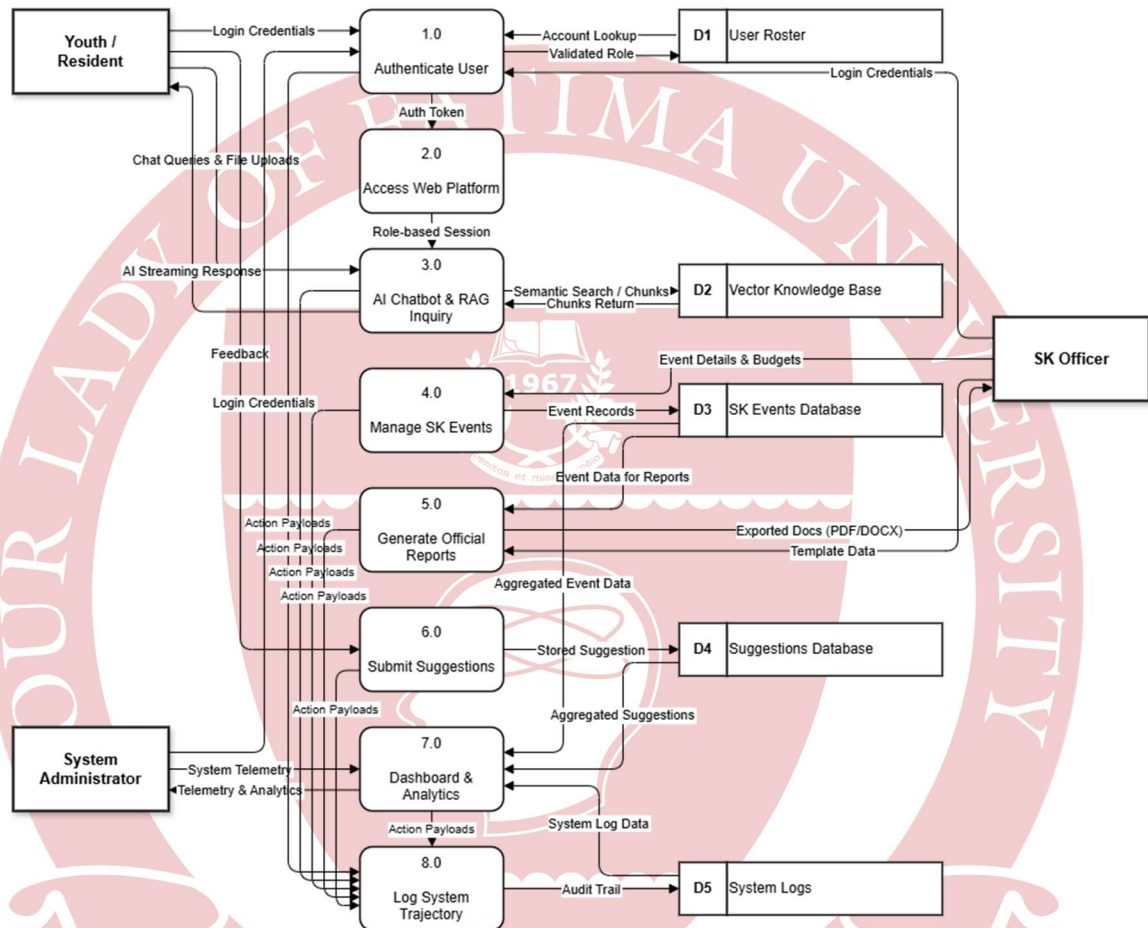


Figure 4. Data Flow Diagram

The figure presents the Data Flow Diagram (DFD) of the proposed aSK Youth system, illustrating how data flows between external entities and the system's core processes. The diagram features three external entities: Youth/Resident, SK Officer, and System Administrator. The process begins when users submit login credentials to the Authenticate User (1.0) module, which performs an account lookup against the D1 User Roster to return

a validated role and auth token, granting role-based session access to the web platform. Youth users can then send inquiries to the AI Chatbot & RAG module, which performs semantic searches against the D2 Vector Knowledge Base to return relevant chunks and responses. SK Officers interact with the D3 SK Events Database to access event details and budgets, while also sending aggregated event data for report generation. The system generates official reports and provides dashboard analytics. Additionally, the system captures action payloads and system telemetry, sending them to the D5 System Logs for audit trail and logging purposes. This DFD demonstrates how authentication, AI-powered assistance, event management, and system logging are seamlessly integrated into a centralized data flow architecture.

System Flowchart

A system flowchart provides a graphical, step-by-step representation of a system's sequential processes. It uses various interconnected symbols to illustrate where the process begins, where it ends, and how data flows between steps. The flowchart also indicates decision points within the system and specifies how those decisions should be handled. Ovals indicate the overall start and end of the process. Arrows link processes and other elements together. Diamonds represent decision points where either the user or the system must choose a path. Rectangles are used for tasks or actions to be performed.

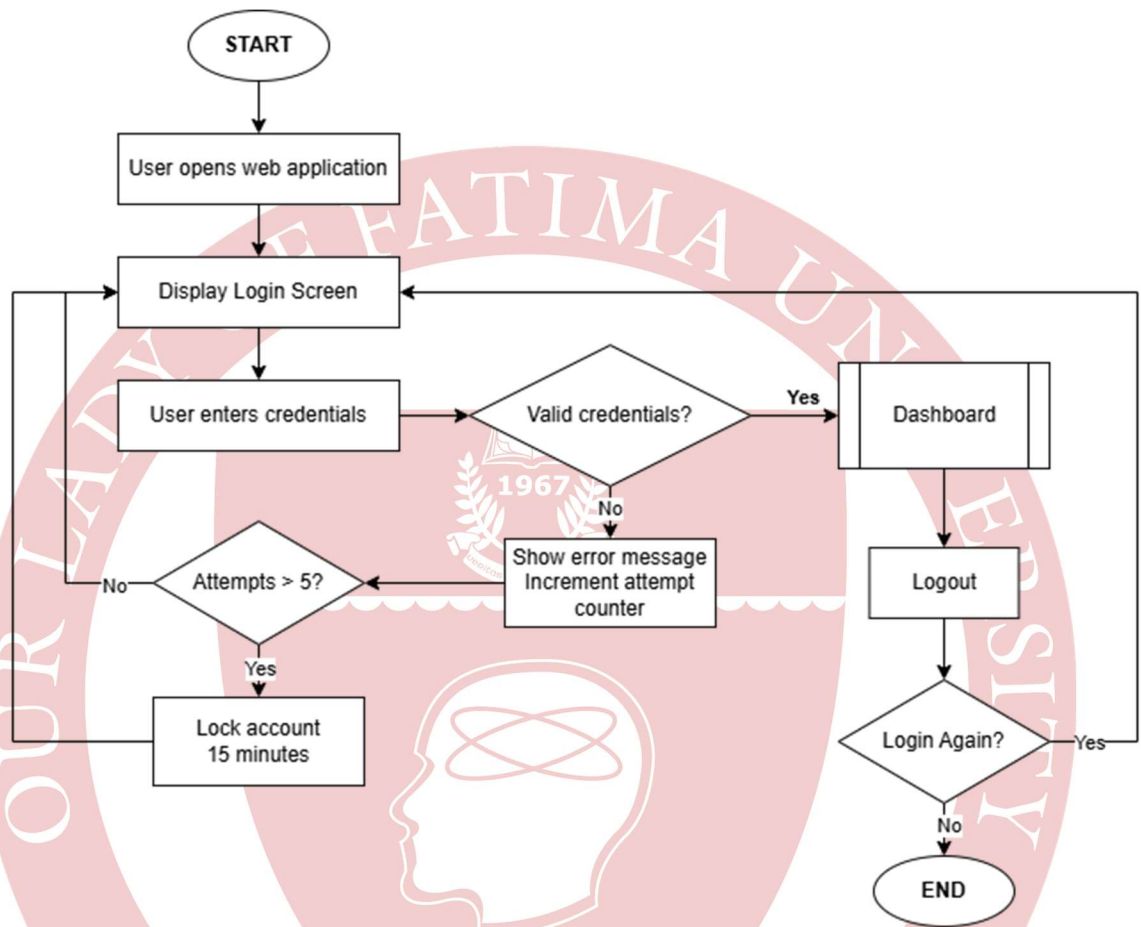


Figure 5a. System Flowchart (Part 1: Authentication & Role Determination)

This system flowchart illustrates the login and role-based access control process of the aSK Youth web application. The process begins when the user opens the web app. The system then checks if the user is already logged in. If not, it displays the login or registration screen where the user enters their credentials then registers. The system verifies whether the credentials are valid; if not, it shows an error message and prompts the user again. Once valid credentials are provided, the system determines the user's role. Based on the role, the

Youth, SK Officer, or Admin, the flow directs the user to their respective dedicated flowchart. This diagram clearly shows how authentication and role-based routing are handled before granting access to role-specific features.

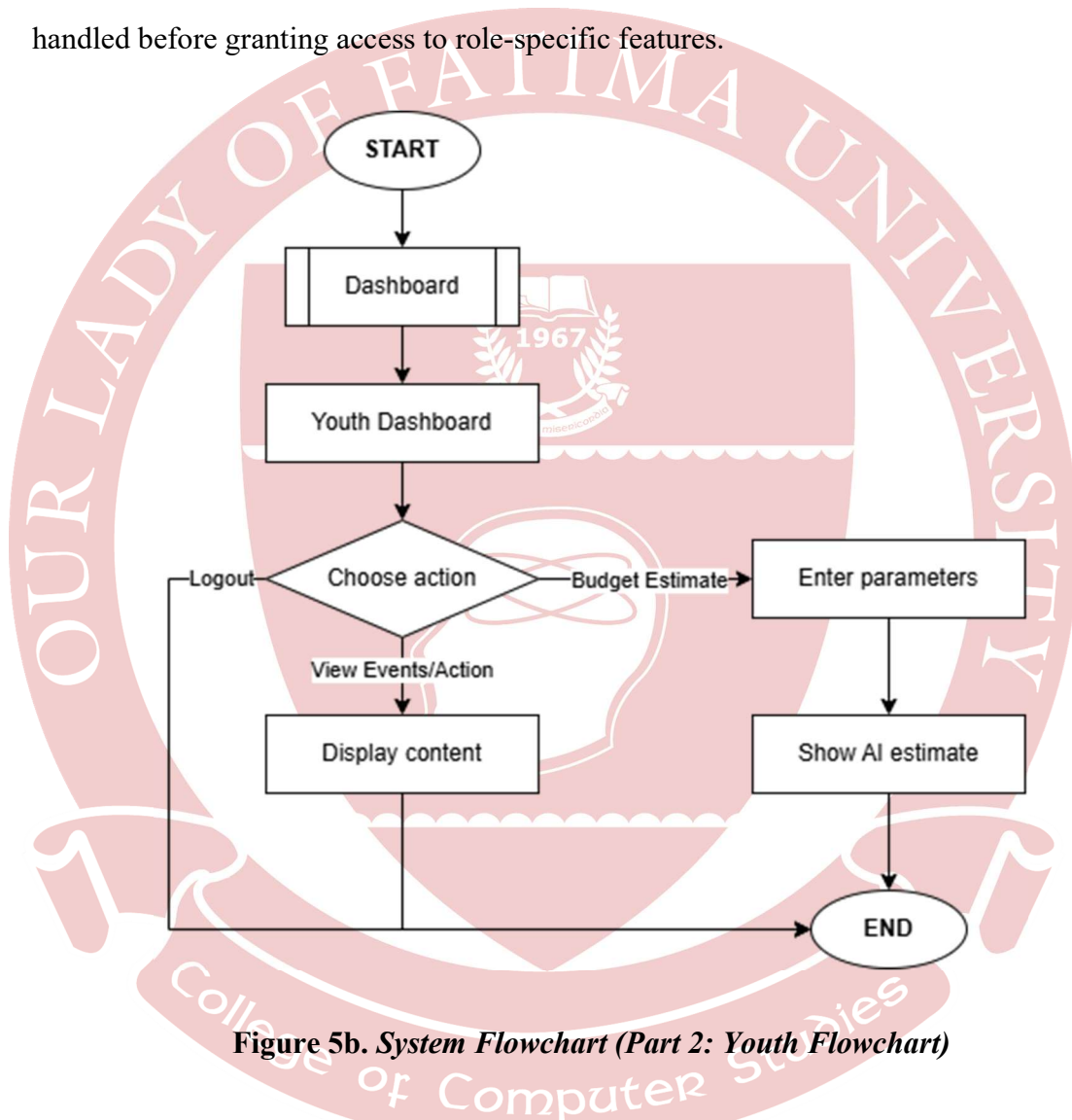


Figure 5b. System Flowchart (Part 2: Youth Flowchart)

Based on the figure above, it represents the flow of the entire login and role-based access process of the aSK Youth web application, which is composed of different symbols as well as the description of every step from user authentication to role-specific redirection.

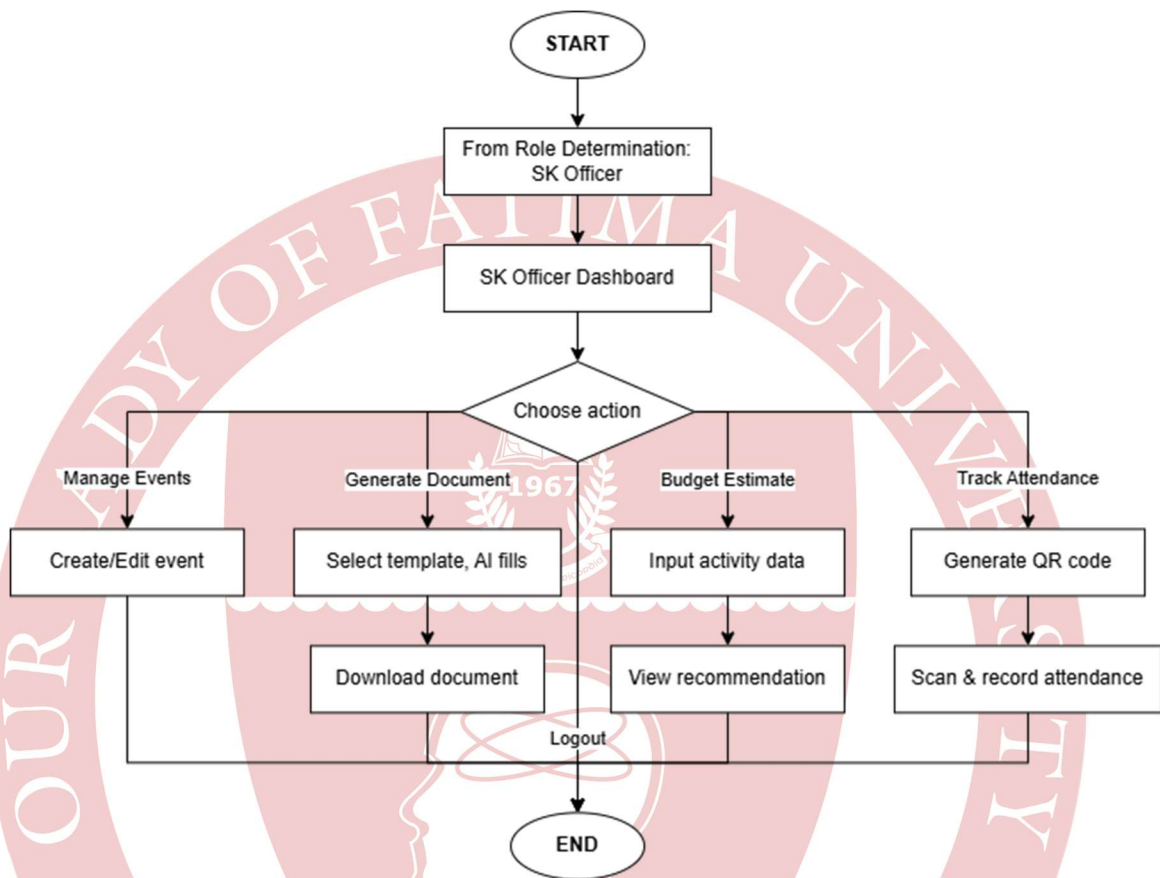


Figure 5c. System Flowchart (Part 3: SK Officer Flowchart)

Based on the figure above, it represents the flow of the SK Officer's process within the aSK Youth web application, which is composed of different symbols as well as the description of every step from accessing the dashboard to managing events, generating AI-filled documents, receiving budget recommendations, tracking attendance via QR code, and logging out.

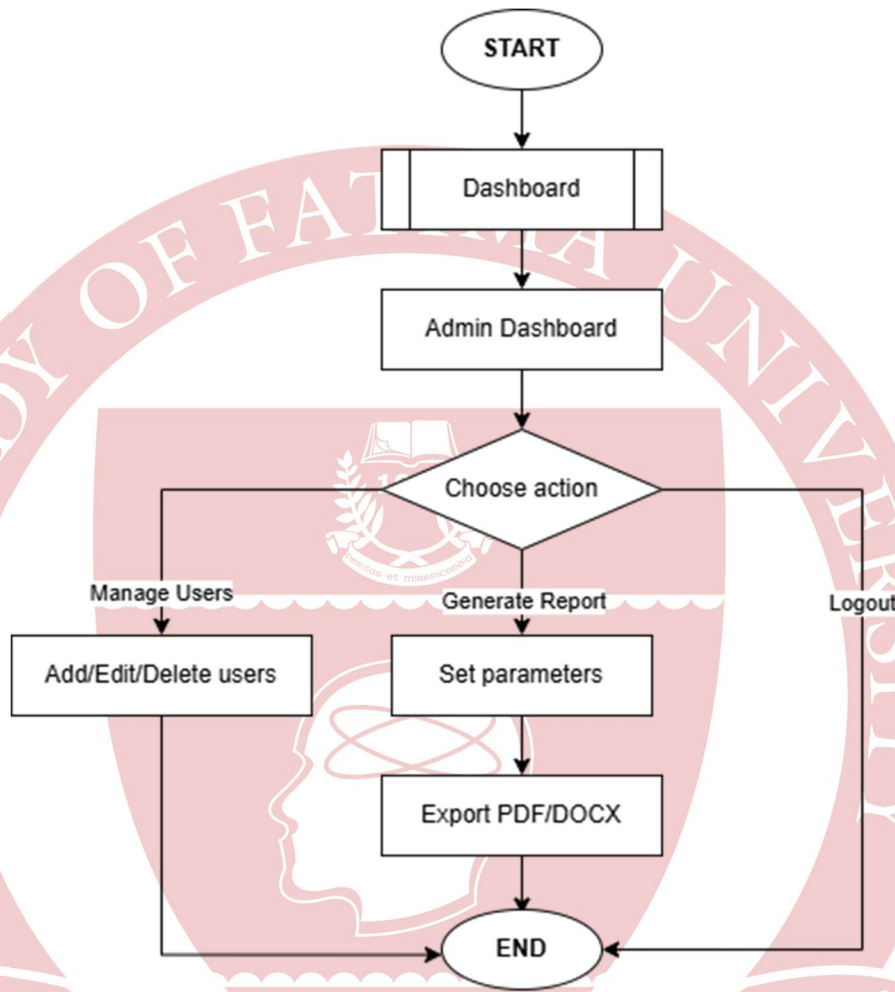


Figure 5d. System Flowchart (Part 4: Admin Flowchart)

Based on the figure above, it represents the flow of the Admin's process within the aSK Youth web application, which is composed of different symbols as well as the description of every step from accessing the dashboard to managing users, generating parameter-based reports exported as PDF or DOCX, and logging out.

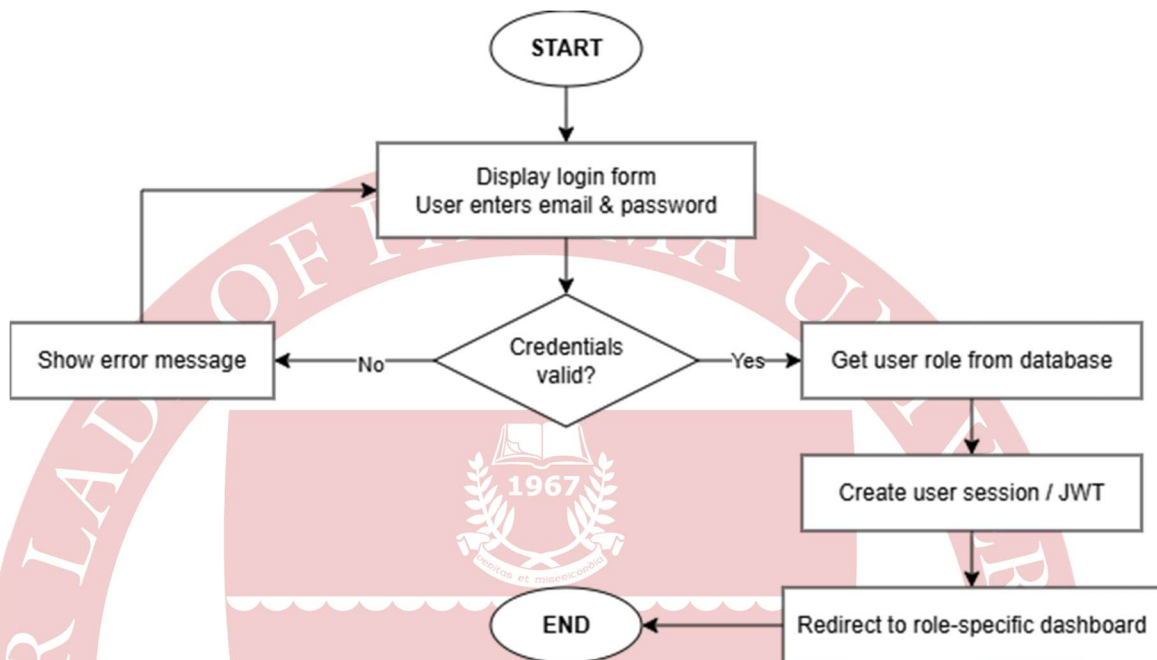


Figure 5e. Program Flowchart (Login Module)

The figure above represents the program flowchart of the login validation and role-based redirection process, which shows how user credentials are verified and how access is granted based on account type. One of the key processes included in the system is the failed attempt tracking and temporary account lockout mechanism. After five consecutive failed password attempts, the account is locked for 15 minutes as a security measure. This process is additional for protecting against brute force attacks.

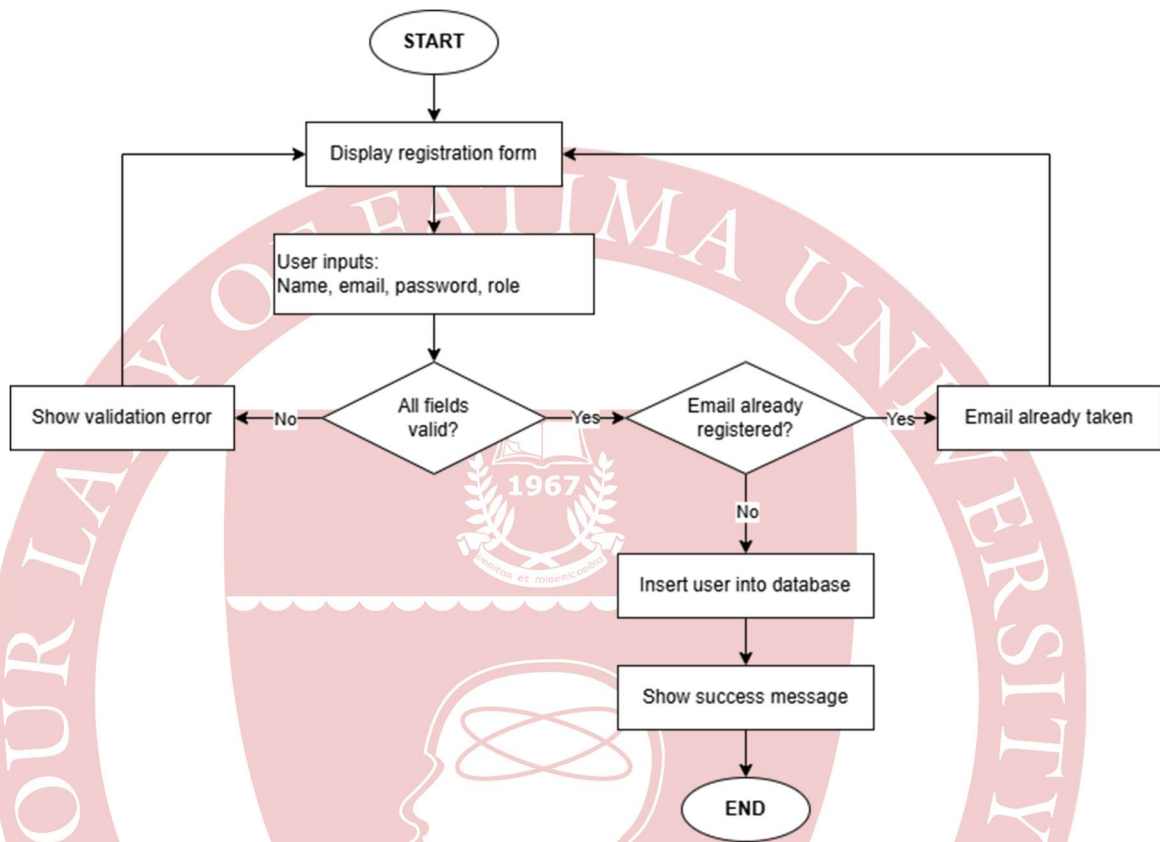


Figure 8f. Program Flowchart (Registration Module)

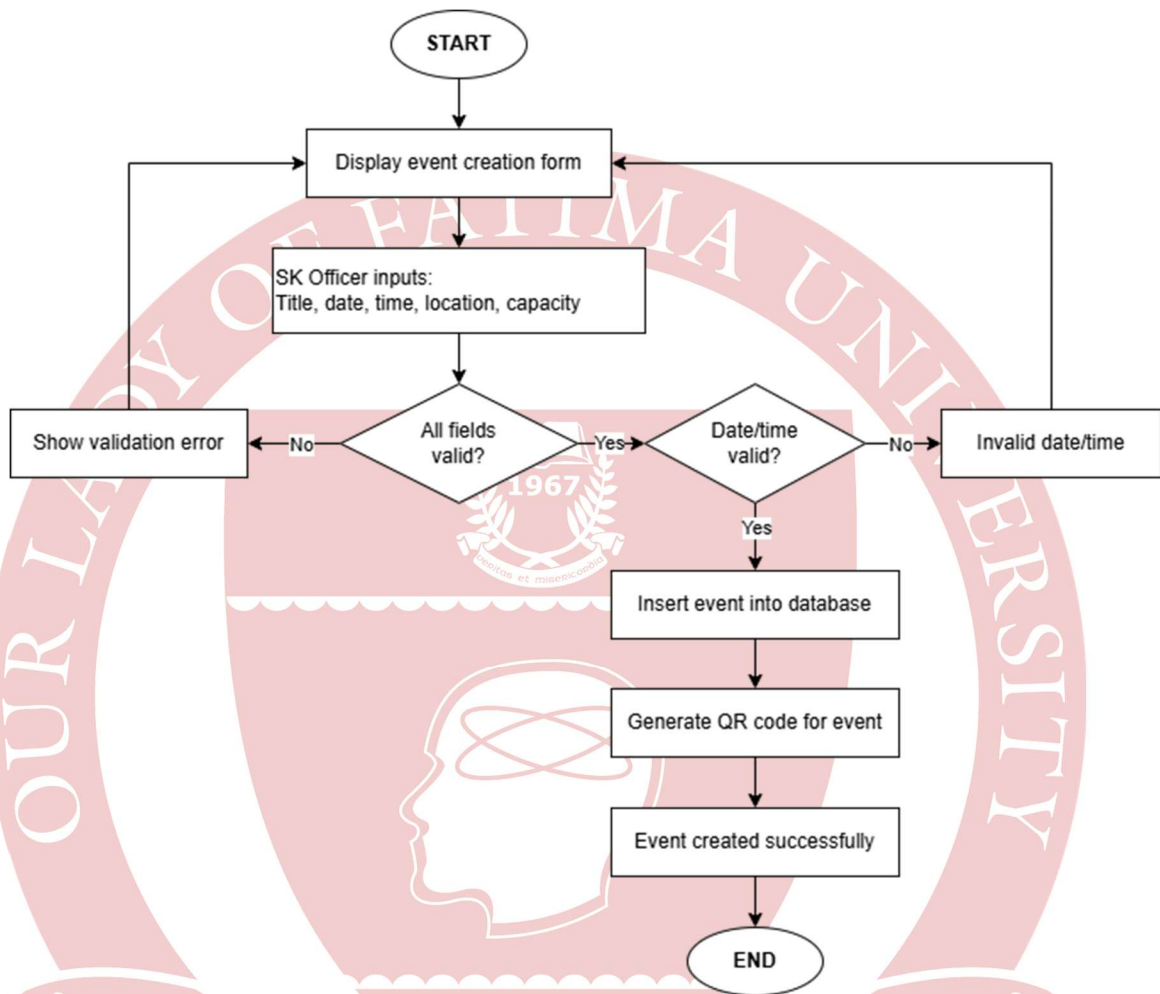


Figure 8g. Program Flowchart (Event Creation Module)

MODULE BREAKDOWN DIAGRAM

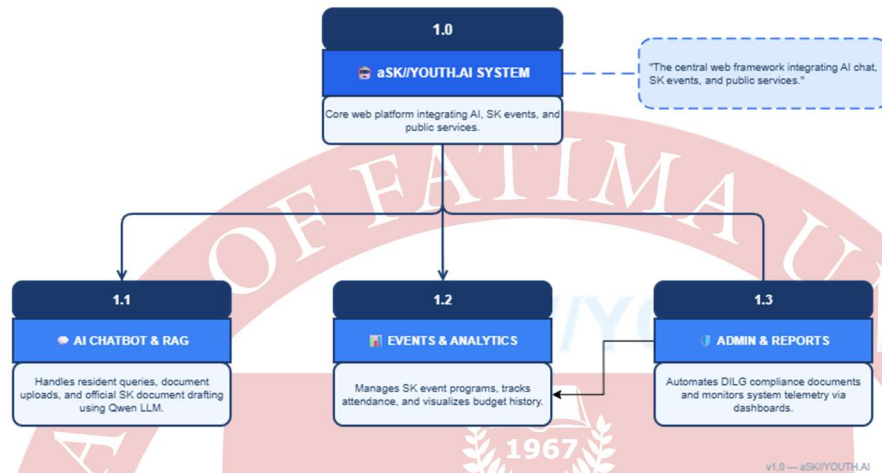


Figure 9. Visual Table of Contents

The Visual Table of Contents outlines the main functional modules organized by user roles and system features. This visual serves as a roadmap for understanding the overall structure and navigation of the aSK Youth System.

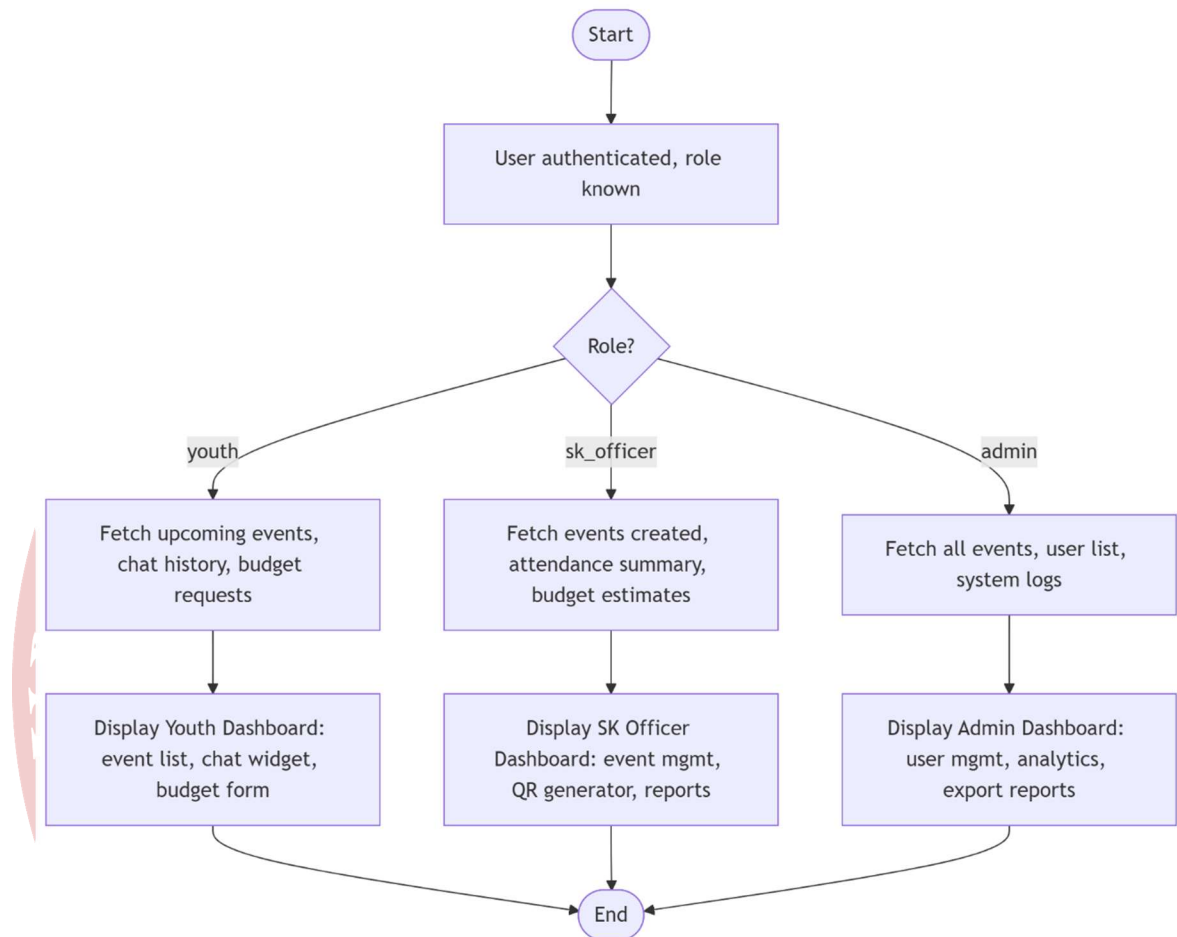


Figure 10. Dashboard Module

The figure shows how the system personalizes the user interface based on each user role after authentication. The process branches into three paths. For youth users, the system fetches upcoming events, chat history, and past budget requests, then displays a dashboard featuring an event list, an AI chat widget, and a budget estimation form. For SK officers, the system retrieves events they have created, attendance summaries, and budget estimates, presenting a dashboard with event management tools, a QR code generator for attendance

tracking, and report options. For administrators, the system pulls comprehensive data including all events, complete user lists, and system logs, displaying a dashboard focused on user management, system analytics, and exportable reports. This role-based approach ensures that each user sees only the tools and information relevant to their responsibilities.

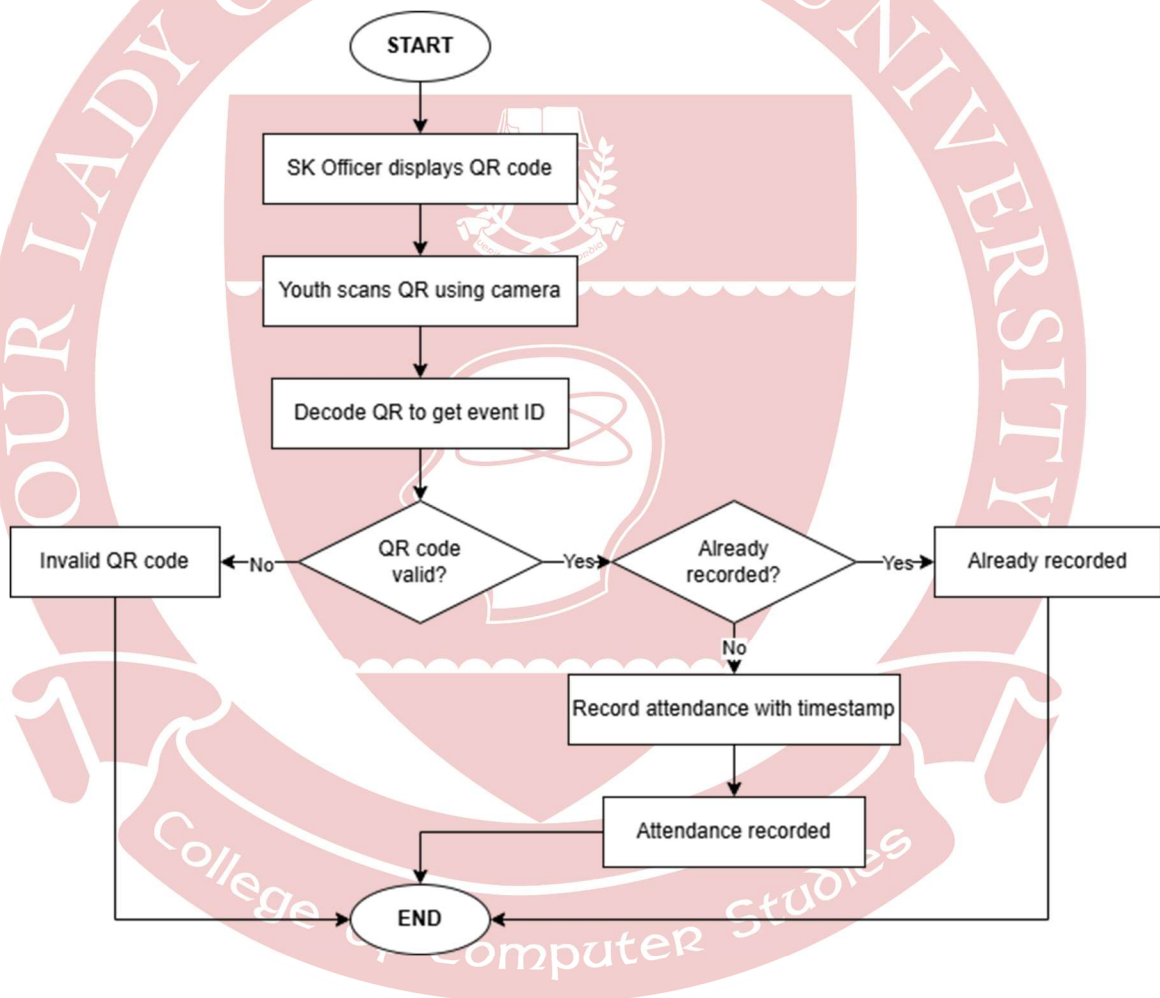


Figure 11. Attendance Module

The figure illustrates the Attendance Module flowchart for the aSK Youth system, which functions through two primary actions: Generate QR and Scan QR. When an SK officer

sets up an event, the system produces a unique QR code that can be printed or shown on a screen. On the participant side, a youth uses their device's camera to scan the QR code, allowing the system to pull the event ID. The system then verifies whether the youth has already been logged for that particular event. If attendance has already been recorded, a message saying "Already recorded" appears. If not, the system logs the attendance with a timestamp, updates the attendee count in the database, and shows a confirmation that the attendance was successfully recorded. This module offers a digital and paper-free way to monitor event participation.

Hierarchical Input Process Output

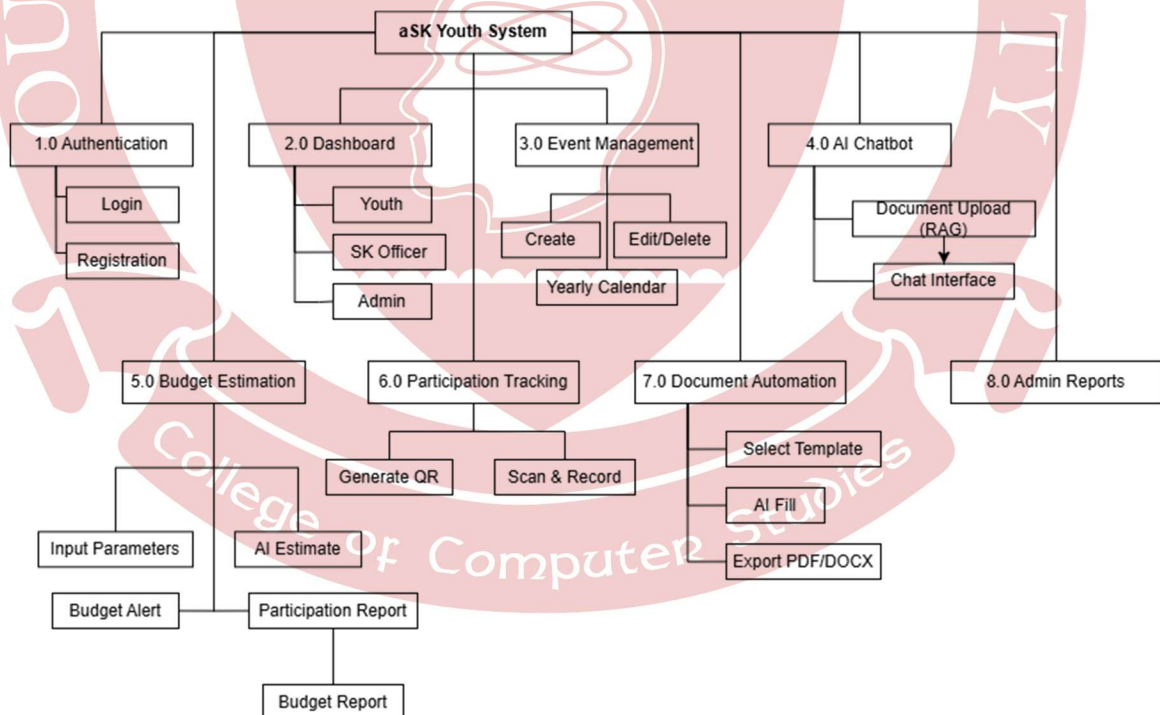


Figure 12. Hierarchical Input Process Output

The figure shows the Hierarchical Input-Process-Output diagram of the aSK Youth system by presenting the structured decomposition of the major system modules and their corresponding functions. It illustrates how system inputs such as login credentials, user roles, event details, chat messages, attendance scans, budget data, and document templates are processed through different modules including Authentication, Dashboard, Events Management, AI Chat with RAG, Budget Estimation, Tracking with QR, Document Templates, and Reports Generation. Each module performs specific processes that transform inputs into meaningful outputs such as role-specific dashboards, created events, AI-generated responses, budget estimates, recorded attendance, exported documents, and printable reports. The hierarchical structure helps in understanding the overall system design using a top-down approach, making the functions of each subsystem clear and organized.

Evaluation Procedures

The evaluation procedures for the aSK Youth system were conducted following the ISO/IEC 25010 software quality standards. This standard provides a framework for assessing the quality and effectiveness of software systems through several important characteristics. The system was evaluated based on the following criteria:

Functionality Suitability: This measures how well the features of aSK Youth meet the needs of the users, particularly in providing youth-related services, announcements,

activity management, and communication within the Sangguniang Kabataan community.

It ensures that the system performs its intended functions accurately and effectively.

Performance Efficiency: This evaluates the system's ability to respond quickly and efficiently while handling multiple users, data processing, and real-time transactions without delays or system interruptions.

Compatibility: This determines the ability of the system to operate smoothly across different devices, browsers, and network environments, ensuring proper communication and accessibility for all users.

Usability: This assesses how easy the system is to learn, navigate, and use for SK officials, youth members, and administrators. It focuses on user-friendliness, accessibility, and overall user satisfaction.

Reliability: This examines the consistency and stability of the system during operation, ensuring that features such as announcements, registrations, and information management function properly with minimal errors or failures.

Security: This evaluates the protection of user data and system information by ensuring that only authorized users can access specific features and that sensitive information is securely stored and managed.

Maintainability: This measures how easily the system can be updated, modified, and improved in the future to accommodate additional features, policy changes, or technical enhancements without affecting system performance.

Portability: This assesses the system's capability to operate on different platforms and devices such as desktop computers, laptops, tablets, and mobile phones, allowing users to access aSK Youth conveniently anytime and anywhere.

Statistical Tools

The following statistical tools will be used to analyze the collected data:

1. Mean – To determine the average rating for each ISO criterion. The formula is:

$$\bar{x} = \frac{\sum x}{n}$$

Where:

- $\bar{x}$ = mean
- $\sum x$ = sum of all scores
- n = number of respondents

The mean will be interpreted using the following scale:

Mean Range

Interpretation

4.51 – 5.00

Strongly Agree

3.51 – 4.50

Moderately Agree

2.51 – 3.50

Satisfied / Neutral

1.51 – 2.50

Moderately Disagree

1.00 – 1.50

Strongly Disagree

2. Standard Deviation – To measure the spread of responses around the mean. The formula is:

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n-1}}$$

3. Paired T-test – To determine whether there is a significant difference between the pre-test (manual system) and post-test (aSK Youth) scores. The formula is:

$$t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{(s_1^2/n_1 + s_2^2/n_2)}}$$



Where:

- $\bar{x}_1$ = mean of pre-test scores
- $\bar{x}_2$ = mean of post-test scores
- s_1^2 = variance of pre-test scores
- s_2^2 = variance of post-test scores
- n_1, n_2 = number of respondents (both 50)

The t-test will be conducted at a 95% confidence level ($\alpha=0.05$). A p-value less than 0.05 indicates a statistically significant improvement.

Ethical Considerations

The study will adhere to the following ethical principles:

- Informed Consent – All respondents will be informed of the purpose of the study and will sign a consent form before participation.

- Confidentiality – Personal information will be anonymized in all reports. Only aggregated data will be published.
- Voluntary Participation – Respondents may withdraw from the study at any time without penalty.
- Data Security – All collected data will be stored on a password-protected server accessible only to the research team.

Requirement Specification: Hardware and Software Specification

To ensure the reliability, accessibility, and efficient performance of the proposed AI-driven SK Barangay Assistant application, the researchers identified the necessary technical requirements for system development, deployment, and end-user operation. The system is primarily designed for desktop and personal computer environments to support administrative and organizational tasks efficiently. In addition, the application is developed with responsive design principles to maintain compatibility across mobile devices and tablets, allowing users to access system features through different platforms when needed. Since the application utilizes cloud-based services for data processing, storage, and AI-powered functionalities, most computational processes are handled remotely. Therefore, the local technical requirements mainly emphasize stable internet connectivity, compatible web browsers, adequate device performance, and responsive user interface functionality to ensure consistent and seamless system operation across supported devices.

Operating System Requirements

- End-user client (browser): Any common desktop or mobile OS that supports a current web browser (e.g. Windows 10 or 11, macOS, Android 10+, iOS / iPadOS current releases).
- Inference and API host (beneficiary or SK-funded PC): 64-bit Windows 10 or 11 is the primary reference environment for this project's launcher scripts and paths; 64-bit Linux is viable for teams comfortable running Node.js, Python, and cloudflared manually with equivalent paths and services.
- Public frontend (Vercel): The built React application (static files after 'vite build') runs on Vercel's managed hosting—not on the student's laptop or on the inference PC. Viewing the site only requires a browser; nobody needs to install Apache, IIS, nginx, or another web server on their own machine to serve the interface. (Operating system updates for Vercel's infrastructure are handled by Vercel.)

Special Software Requirements

- End-user client: A modern web browser (Google Chrome, Microsoft Edge, Mozilla Firefox, or Apple Safari, current stable channel) with JavaScript enabled, support for 'fetch' and readable streams for chat, and optional PDF viewing for exported documents or reports where the application provides downloads.

- Inference and API host: Node.js (LTS recommended) for the Express backend and node-llama-cpp; Python 3 with FastAPI / Uvicorn for the companion AI layer on port 8000; NVIDIA GPU drivers and a CUDA-capable stack for GGUF inference on GPU; Cloudflare Tunnel connector (cloudflared) with tunnel token or credentials so `https://` API hostnames forward to `localhost:3001`; optional Tesseract / image tooling where OCR features are enabled. Embedded SQLite (via better-sqlite3) satisfies persistence—no separate MySQL or PostgreSQL server is required for this codebase.
- Cloud services: Vercel account and project for building and hosting the Vite + React client from `'askyouth-web-only'`.

Hardware and Cloud Specifications

Hardware requirements distinguish lightweight clients that only render the hosted UI from the inference PC that must sustain the model, vector store, and tunnel.

- End-user client: Dual-core (or better) processor, 4 GB RAM minimum (8 GB recommended for smooth operation with charts and long threads), stable Wi-Fi or Ethernet, and a display suitable for reading and typing; tablets and phones should meet their OS vendor's minimums for current browsers.
- Public frontend (Vercel): Performance and global delivery are provided by Vercel's edge network; no hardware is consumed for hosting static assets.

- Inference and API host (beneficiary or SK-funded PC): x86-64 CPU with eight physical cores or sixteen threads (or equivalent) recommended so Node, Python (transformers / PyTorch on CPU per project design), and OS stay responsive; 32 GB system RAM recommended for concurrent services and headroom; NVMe or SSD with tens of gigabytes free for application files, dependencies,
- GGUF weights (multi-GB for Qwen 2.5 7B quantized), embedding caches, and SQLite plus vector data—exact size depends on installed models and retention, not on whether a developer machine keeps duplicate frontend copies.
- GPU (inference host): NVIDIA discrete GPU with CUDA support and minimum 8 GB VRAM for the intended Qwen 2.5 7B Instruct path in this stack; AMD or Intel-only GPUs do not satisfy the current CUDA inference configuration without changing the stack. A reference-class workstation similar to AMD Ryzen 7 7700, 32 GB DDR5, and NVIDIA GeForce RTX 4060 Ti 8 GB illustrates a configuration that meets the 8 GB VRAM floor; larger context or future models may warrant more VRAM.
- Network (inference host): Broadband with at least ~10 Mbps upload and ~10 Mbps download recommended so Cloudflare Tunnel stays stable, health checks succeed, and document uploads complete without timeouts.

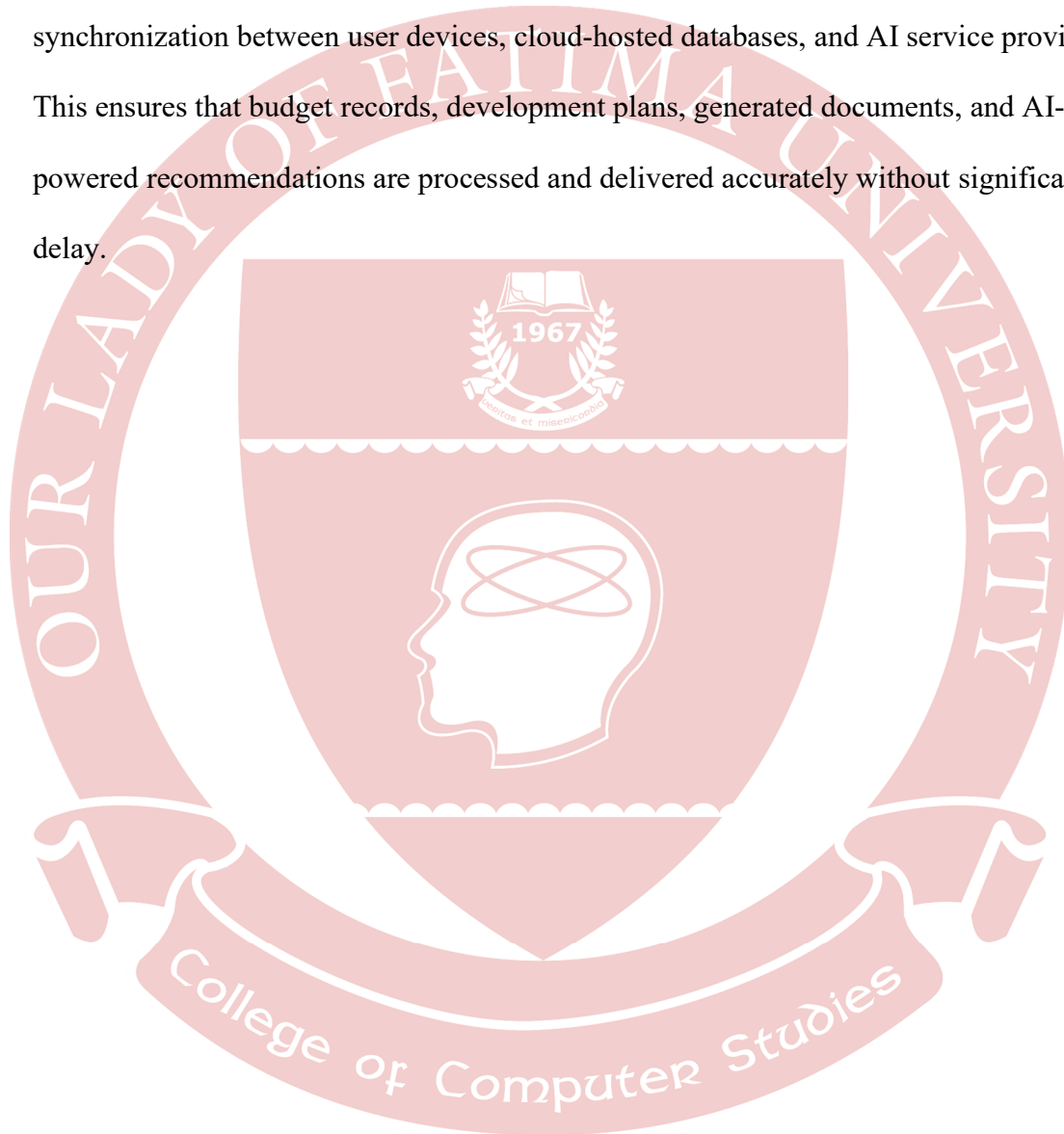
Network Requirements.

To ensure the reliable operation, accessibility, and real-time performance of the proposed AI-driven SK Barangay Assistant application, a stable internet connection is required for all supported devices. Since the system utilizes cloud-based infrastructure, remote database services, and AI-powered processing, continuous network connectivity is essential for seamless data transmission, system synchronization, and user interaction.

A stable internet connection with a minimum bandwidth of 10 Mbps through Wi-Fi or wired Ethernet connection is recommended for desktop and laptop users. While bandwidth capacity may vary depending on usage, consistent network stability is essential as the system relies on remote cloud servers for budget retrieval, document generation, bilingual processing, and AI-assisted planning functionalities.

Mobile and Tablet Devices: Mobile devices and tablets require a reliable Wi-Fi or mobile data connection to access the web-based interface, retrieve organizational records, submit user requests, and receive AI-generated recommendations in real time. This ensures continuous accessibility for SK officials who may need remote access to system features outside standard workstation environments.

External Network Connectivity: A dependable Internet Service Provider (ISP) with stable connectivity and minimal network latency is required to support real-time synchronization between user devices, cloud-hosted databases, and AI service providers. This ensures that budget records, development plans, generated documents, and AI-powered recommendations are processed and delivered accurately without significant delay.



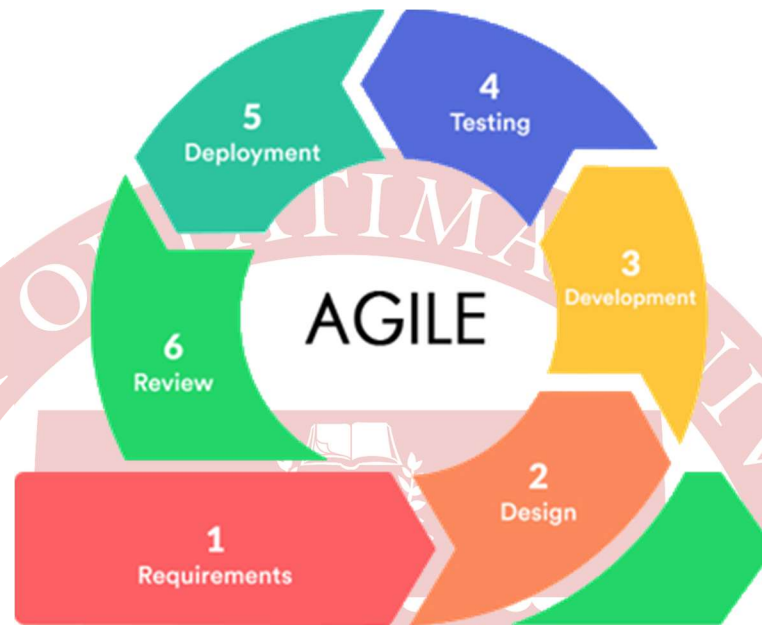


Figure 3.3 Agile Development Methodology

The proponents will use the Agile model, which consists of six iterative phases:

- Requirements – Gather functional and non-functional requirements through interviews with SK officials and youth constituents of Barangay Concepcion Dos.
- Design – Create system architecture, database schema, UML diagrams (use case, sequence, class, activity), and user interface prototypes.
- Development – Implement the system modules using React.js (frontend), Node.js (backend), Python (AI microservices), and a fine-tuned LLM for the chatbot.
- Testing – Conduct unit testing, integration testing, and alpha testing by the development team.
- Deployment – Deploy the system on a local server at Barangay Concepcion Dos.

- Review – Conduct beta testing with end-users and evaluate the system using ISO 25010.

Respondents of the Study

The respondents will be divided into two groups:

Group	Description	Number of Respondents
SK Officials	Elected SK officials and SK committee members of Barangay Concepcion Dos	20
Youth Constituents	Registered youth (aged 18–24) residing in Barangay Concepcion Dos	30
Total		50

The sample size will be determined using Slovin's formula:

$$n = \frac{N}{1 + Ne^2}$$

Where:

- n = sample size
- N = population size (estimated 150 youth stakeholders in Barangay Concepcion Dos)
- e = margin of error (0.05)

Computation:

$$n = \frac{150}{1 + 150(0.05)^2} = \frac{150}{1 + 150(0.0025)} = \frac{150}{1 + 0.375} = \frac{150}{1.375} \approx 109$$

To ensure feasibility, the proponents will target 50 respondents (20 SK officials + 30 youth constituents) using purposive sampling.

Data Gathering Instrument

Three instruments will be used:

1. Semi-structured Interview Guide – Used to collect qualitative data on existing SK workflows, challenges, and feature requirements from SK officials. Sample questions:

- “What are the most time-consuming administrative tasks you perform?”
- “How do you currently track youth participation in events?”
- “What challenges do you face in preparing project budgets?”

2. ISO/IEC 25010-based Survey Questionnaire – A 40-item questionnaire measuring eight quality characteristics. Each item uses a 5-point Likert scale:

- 5 = Strongly Agree
- 4 = Agree
- 3 = Neutral
- 2 = Disagree
- 1 = Strongly Disagree

The questionnaire will be administered twice: as a pre-test (evaluating the existing manual system) and as a post-test (evaluating the aSK Youth system after deployment).

3. **System Usability Log** – Automated logs will record task completion times (e.g., document generation time, budget estimation time) and error rates for quantitative performance analysis.

Data Gathering Procedure

The study will follow these steps:

Phase	Activity	Duration
1	Secure approval from the Barangay Captain and SK Chairperson of Concepcion Dos.	-
2	Conduct interviews with SK officials to document current workflows and requirements.	-
3	Administer the pre-test ISO 25010 questionnaire to all respondents (evaluating the manual system).	-
4	Develop the aSK Youth system following Agile methodology.	-

5 Deploy the system on a local server at the barangay hall and -
conduct user training.

6 Allow respondents to use the system for one month to manage -
actual SK activities.

7 Administer the post-test ISO 25010 questionnaire to the same -
respondents.

8 Export system usability logs for analysis. -

9 Analyze data using statistical tools. -
